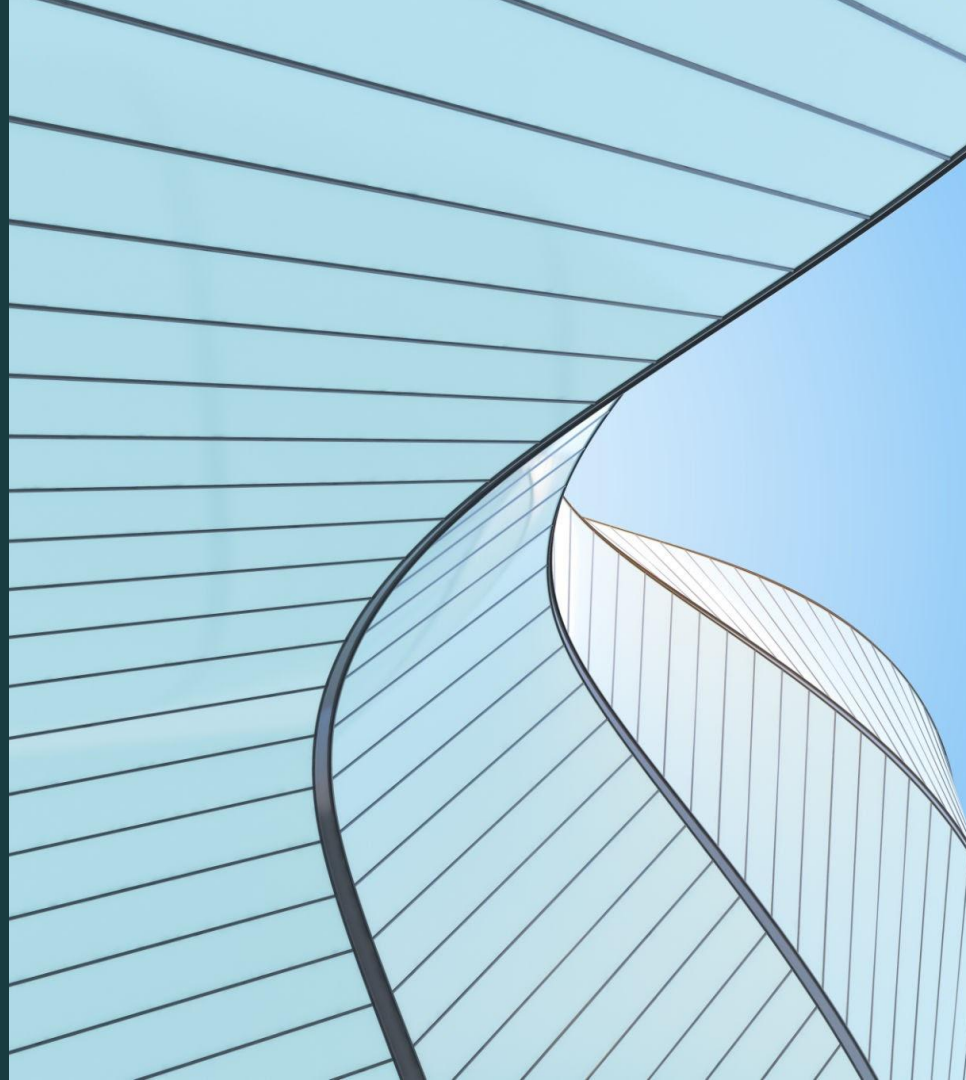
DSI - Team 2
March 7, 2026

DSI Showcase: Bank Marketing

Audience: Bank Executives, Internal Marketing Team



Content *overview*

1

Our Mission & The Problem

2

Dataset Overview: Variables *and* Risks, Unknowns & Biases

3

Our Process & Tools Used

4

Key Insight

5

Business Impact

6

Future Work

Our *team*

Kateryna Makieieva

TITLE

name@example.com

Kerensa Wong

kerensa.ca@gmail.com

Cookiejars8

TITLE

name@example.com

Vince Nguyen

TITLE

name@example.com

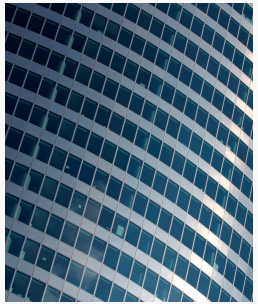
Full Name

TITLE

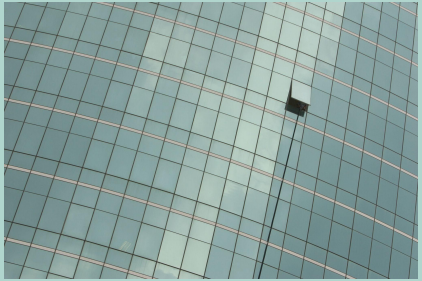
name@example.com

OUR MISSION

Optimize bank marketing campaigns by targeting the right customers to increase subscription rates and reduce wasted outreach.



This project focuses on improving direct marketing efficiency using data-driven insights. By identifying high-probability customers, the bank can allocate resources more effectively, reduce unnecessary calls, and improve overall campaign performance. The goal is to replace broad outreach with targeted strategy.



THE PROBLEM

Banks often rely on mass-calling strategies for term deposit campaigns. This results in:

- Low conversion rates
- High operational costs
- Agent time spent on low-probability customers
- Customer fatigue from excessive follow-ups

The gap: There is no optimized targeting strategy based on customer history, financial profile, and campaign behavior.

DATASET OVERVIEW: VARIABLES

The dataset comes from the UCI Bank Marketing dataset contains multi-year campaign data; approximately 45,000 structured customer records from direct marketing campaigns, *with a binary target variable y indicating whether a client subscribed to a term deposit (yes/no).*

Client Data (Customer Profile)

Basic customer background characteristics used for segmentation:

- age
- job
- marital status
- education

Financial Profile

Customer financial condition and credit obligations:

- balance (account balance)
- housing (loan status)
- loan (personal loan status)
- default (credit in default)

Campaign Data

Information related to the current marketing campaign contact:

- contact (cell or tele)
- last contact month
- call duration in secs.
- number of contacts during current campaign

Previous Campaign Data

Customer interaction history from prior campaigns:

- pdays (days since last contact)
- previous (number of previous contacts)
- poutcome (outcome of previous campaign)

DATASET OVERVIEW: RISKS, UNKNOWNS & BIASES

Write a *high-level insight* about the competitive ecosystem

	1	2	3	4
RISKS	• List the features of this product or service • 2 or more	• List the features of this product or service • 2 or more	• List the features of this product or service • 2 or more	• List the features of this product or service • 2 or more
UNKNOWNS	• Point out the strengths of this product or service • 2 or more	• Point out the strengths of this product or service • 2 or more	• Point out the strengths of this product or service • 2 or more	• Point out the strengths of this product or service • 2 or more
BIASES	• Call out the weaknesses of this product or service • 2 or more	• Call out the weaknesses of this product or service • 2 or more	• Call out the weaknesses of this product or service • 2 or more	• Call out the weaknesses of this product or service • 2 or more

OUR PROCESS

Applying structured analysis to transform raw customer data into targeted marketing strategy.

01
DATA CLEANING

Prepared and structured the dataset for analysis.

- Checked for missing or inconsistent values
- Standardized categorical variables
- Verified data types and distributions
- Removed or reviewed extreme outliers

02
EDA

Identified patterns and relationships within the data.

- Analyzed subscription rates across segments
- Compared demographic and financial groups
- Evaluated balance distributions
- Reviewed campaign variable behavior

03
ANALYSIS

Measured the impact of key variables on subscription outcomes.

- Compared conversion rates by previous campaign outcome
- Evaluated effect of call duration and contact frequency
- Assessed impact of financial stability (balance, loans, default)
- Analyzed performance by campaign month

04
VALUE PROPOSITION

Focused on actionable business insights rather than model complexity.

- Identified high-probability customer segments
- Determined optimal campaign timing
- Recommended efficient follow-up strategy
- Prioritized financially stable customer profiles

TOOLS LEVERAGED

Python	Data cleaning, analysis, and exploration
Pandas & NumPy	Data manipulation and preprocessing
Matplotlib / Seaborn	Data visualization and pattern identification
Streamlit	Interactive data presentation and deployment
Jupyter Notebook	Development and analysis environment

KEY INSIGHTS

Streamlit Dashboard

BUSINESS IMPACT

Announce your vision for scaling operations

STEP TITLE

Define your first goal toward making your business more productive and efficient.

01

STEP TITLE

Identify what you'll focus on next to help your team work together and grow your operations.

02

STEP TITLE

Describe the outcome you want to achieve next. Provide any relevant details about how you'll get there.

03

STEP TITLE

Bring it all together. Share how you'll evaluate, maintain, and advance your operations for long-term sustainability.

04

BUSINESS IMPACT

Evaluate your revenue trajectory relative to your goals

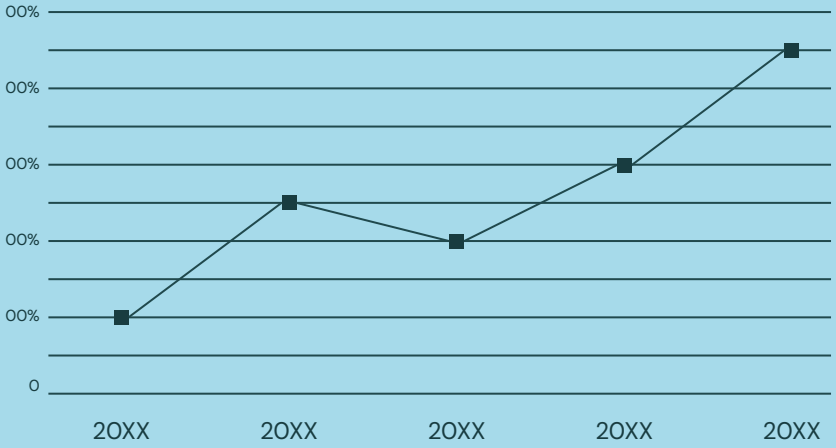
Yearly revenue

$00

Net revenue

Gross revenue

Revenue growth



FUTURE WORK

From a PM perspective I would utilise this section to put anything the team couldn't achieve in the given time/resources for this project, or any ideas the team has regarding improving the value for the selected stakeholders given the chosen project context

Key learning 1

Go deeper into a specific insight from the competitive analysis and what it means for your business.

Key learning 2

Point to the gap your product will fill and explain how your business can claim that space.

Key learning 3

Consider your brand and customer segments. Explain why you're ideally positioned to reach them.

Thank you

Conclude the presentation by thanking your audience for listening and participating.

Key Insights - Bu

Context

Provide some insight into the state of the market today. Describe any challenges, forces, or factors influencing the market. Share your company's perspective on where the market has been and where it's headed.

Market value	00
Historic growth rate	00
Projected growth rate	00
Cost of entry	00

Key players

Customer segments

Segment 1	Segment 2
Segment 3	Segment 4

Pricing strategy

	PRICE POINT		
MARKET	High	Medium	Low
Premium			
Mid-level			
Economy			

Key Insights - Customer

Key players

Competitor brand

Competitor brand

Competitor brand

Competitor brand

Competitor brand

Competitor brand

Market value	00
Historic growth rate	00
Projected growth rate	00
Cost of entry	00

Context

Provide some insight on the state of the market today. Describe any challenges, forces, or factors influencing the market. Share your company's perspective on where the market has been and where it's headed.

Customer segments

Segment 1

Segment 2

Segment 3

Segment 4

Pricing strategy

	PRICE POINT		
MARKET	High	Medium	Low
Premium			
Mid-level			
Economy			

THE OPPORTUNITY

Based on your *competitive analysis* in previous slides, describe the *growth opportunities for your business*. Write a comprehensive claim here, then break it down into individual learnings.

Key learning 1

Go deeper into a specific insight from the competitive analysis and what it means for your business.

Key learning 2

Point to the gap your product will fill and explain how your business can claim that space.

Key learning 3

Consider your brand and customer segments. Explain why you're ideally positioned to reach them.

Company Name

MARKET ANALYSIS

Provide *key insights* into the state of the market. Share your company's perspective on where the market has been and where it's headed.

Add any additional context about forces or factors influencing the market and how your business is meeting the moment.

00%

Add a few words to explain this metric.

00M

00+

Add a few words to explain this metric.

Add a few words to explain this metric.

00M

Add a few words to explain this metric.

BUSINESS MODEL

Clearly outline your business model. This is how you'll create, deliver, and capture value.

Cost structure

What are the major costs of your proposal? Think about salaries, rent, or marketing.

- Expense 1
- Expense 2
- Expense 3

Revenue streams

Outline how you plan to generate revenue through sales, subscriptions, or other methods.

- Revenue 1
- Revenue 2
- Revenue 3

Key activities

What are the core activities your company undertakes to deliver its value proposition? Think about production, research, or fabrication.

Key partnership

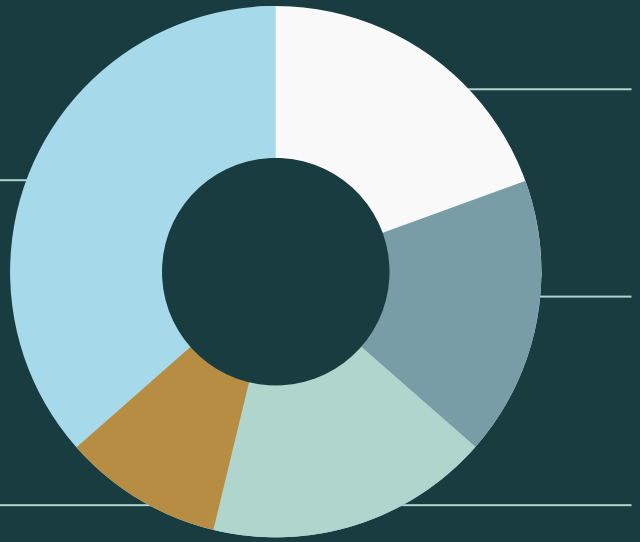
Who are your key partners or suppliers? What resources do they provide? And how do they contribute to your business model?

Customer segments

Use this space to restate or prioritize specific customer segments based on your business model.

- Customer segment
- Customer segment
- Customer segment

COST STRUCTURE



- Staffing
- Taxes
- Marketing
- Logistics
- Technology

-00%

Estimate potential cost savings in an area or department.

-00%

Add any other potential cost savings in an area or department

If you anticipate changes to your expenses, mention the impact to your cost structure here.

REVENUE MODEL

00%

Revenue stream 1

Define how you'll generate income from this source. Identify relevant pricing or channels.

00%

Revenue stream 2

Define how you'll generate income from this source. Identify relevant pricing or channels.

00%

Revenue stream 3

Define how you'll generate income from this source. Identify relevant pricing or channels.

00%

Revenue stream 4

Define how you'll generate income from this source. Identify relevant pricing or channels.

00%

Revenue stream 5

Define how you'll generate income from this source. Identify relevant pricing or channels.

KEY ACTIVITIES

Primary activity *name*

Describe the core activity of your business, how you create and deliver your products or services. This might be manufacturing, product design, software engineering, customer service, marketing, or project management.

Support activity name

Break down your primary activity into the additional activities required to successfully deliver your value proposition. These should contribute to your core business function.

Support activity name

Think about your everyday operations. For example, if your primary activity is software engineering, your support activities may be customer support, software integration, or maintenance.

Support activity name

Show your audience what a day in the life of your organization looks like. Help them understand what's required to do what you do.

CUSTOMER SEGMENTS

Define your target audiences and how you'll speak to them

Target audience 1

Describe a target audience or customer group you want to reach. Identify their traits, values, behaviors, or preferences.

Target audience 2

While your user personas represent ideal individual customers, segments represent a wider range of people who share certain characteristics.

Target audience 3

Think about how each target audience or segment relates to your product or service, why they need it, and how they'll use it.

Target audience 4

Refer back to your user personas and point out which segments they belong to. This will help your presentation feel more interconnected.

Target audience 5

For example, a tech-savvy user persona might fall into a broader Tech Enthusiast or Industry Professional audience segment.

NEXT STEPS

Invite your audience to *learn more* about your organization and to *keep in touch*.

Contact
Full Name
Title
name@example.com

example.com